

Exercise 3.8 (g)

Show that the Takagi function $T(x) = \sum_{i=0}^{\infty} 2^{-i} h(2^i x)$ is not differentiable at any x .

Solution. Let $u_n = \frac{\lfloor 2^n x \rfloor}{2^n}$ and $v_n = u_n + 2^{-n}$. Then $u_n \uparrow x$, $v_n \downarrow x$, and

$$\frac{T(v_n) - T(u_n)}{v_n - u_n} = \sum_{i=0}^{n-1} \frac{h_i(v_n) - h_i(u_n)}{v_n - u_n}, \quad h_i(t) = 2^{-i} h(2^i t).$$

Each term equals ± 1 , so the partial sums do not converge. If $T'(x)$ existed, these difference quotients would converge to $T'(x)$, a contradiction. Hence T is nowhere differentiable.

Exercise 3.9 (*)

Suppose $F = (f_i)_{i=0}^{\infty}$ is a family of continuous functions $f_i : [a, b] \rightarrow \mathbb{R}$, each differentiable on (a, b) , and assume

$$\sup_{x \in (a, b)} |f'_i(x)| \leq M \quad \text{for all } i,$$

for some M independent of i . Show that F is uniformly equicontinuous.

Solution. By the mean value theorem, for any $x, y \in [a, b]$ with $x \neq y$ there exists $c \in (x, y)$ such that

$$|f_i(x) - f_i(y)| = |f'_i(c)| |x - y| \leq M |x - y|, \quad \text{for every } i.$$

Thus each f_i is Lipschitz with the same constant M . Given $\varepsilon > 0$, choose $\delta = \varepsilon/M$. Then for all i and all $x, y \in [a, b]$, if $|x - y| < \delta$ we have $|f_i(x) - f_i(y)| < \varepsilon$. Hence F is uniformly equicontinuous on $[a, b]$.